

Week 4 Report

Executive Summary

This report presents a comprehensive Security Operations Center (SOC) project demonstrating real-world capabilities across threat detection, response, automation, and continuous improvement. The project simulates a full attack lifecycle including initial compromise, lateral movement, and exploitation using controlled environments. Detection is achieved using SIEM tools, followed by alert triage and automated response using SOAR. Adversary emulation validates detection logic, while post-incident analysis identifies root causes and process gaps. Key performance metrics such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and dwell time are calculated to evaluate SOC efficiency. The findings highlight the importance of proactive threat hunting, automation, and structured response workflows. Recommendations include improving detection rules, enhancing playbooks, and strengthening user awareness. This project reflects industry-standard SOC practices aligned with MITRE ATT&CK; and NIST frameworks, making it suitable for professional portfolio and real-world SOC readiness.

1. Threat Hunting Methodologies

A hypothesis-driven threat hunting exercise was conducted to identify unauthorized privilege escalation activities. Logs were analyzed for Event ID 4672 indicating special privilege assignments. Suspicious activity was detected where a non-administrative user gained elevated privileges unexpectedly. Cross-referencing with threat intelligence sources revealed potential misuse aligned with MITRE ATT&CK; technique T1078 (Valid Accounts). Further investigation confirmed anomalous login patterns from unusual IP addresses. The findings indicate compromised credentials and highlight the need for stricter access controls and monitoring. This hunt demonstrates the effectiveness of proactive detection in identifying hidden threats before escalation.

Core Concepts

- **Proactive Threat Hunting** focuses on identifying anomalies such as privilege escalation using Event ID 4672.
- **Hunting Frameworks** like SqRR and TaHiTI provide structured methodologies for hunting.
- **Data Sources for Hunting** include EDR logs, SIEM alerts, network traffic, and threat intelligence feeds.

Timestamp	User	Event ID	Notes
2025-08-18 15:00	testuser	4672	Unexpected admin role

2. Advanced SOAR Automation

A SOAR playbook was developed to automate phishing incident responses. It checks IP reputation, blocks malicious IPs, and creates incident tickets. Integration with SIEM and threat intelligence ensures rapid response, reducing manual workload and improving SOC efficiency.

Step	Status	Notes
Check IP	Success	Malicious
Block IP	Success	Blocked attacker IP 192.168.1.102

- **Alert Triage & Validation**

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download	192.168.1.102	High	Open

Alert was validated using threat intelligence tools. File hash confirmed malicious, triggering escalation.

- **Evidence Analysis**

Item	Description	Collected By	Date	Hash
Log	Server-Z	SOC Analyst	2025-08-18	SHA256

3. Post-Incident Analysis and Continuous Improvement

Root Cause Analysis identifies underlying causes using methods like 5 Whys and Fishbone Diagram.

Question	Answer
Why attack succeeded?	User clicked phishing link
Why link clicked?	Weak filtering

4. Adversary Emulation Techniques

Adversary emulation was conducted using controlled attack simulation techniques to replicate phishing (T1566) and remote exploitation (T1210). Detection systems successfully identified phishing attempts but showed delays in detecting lateral movement. Alerts were generated in SIEM, validating detection rules. However, gaps were identified in response timing and alert correlation. Recommendations include improving detection signatures and automating response workflows. This exercise demonstrates the importance of continuous testing and validation of SOC capabilities against evolving threats.

Timestamp	TTP	Detection	Notes
2025-08-18	T1566	Detected	Email blocked

5. Root Cause Analysis

Question	Answer
Why attack succeeded?	User clicked phishing link
Why link clicked?	Weak email filtering
Why is filtering weak?	No advanced detection
Why no detection?	Misconfigured rules
Why misconfigured?	Lack of review process

6. SOC Metrics

Metric	Value
MTTD	2 Hours
MTTR	4 Hours
Dwell Time	6 Hours

7. Capstone Incident Report

A simulated cyber-attack was conducted using a vulnerable environment to demonstrate end-to-end SOC operations. The attack began with exploitation of a known vulnerability, allowing unauthorized access to the system. The attacker attempted lateral movement and privilege escalation. Detection was achieved through SIEM alerts triggered by abnormal behavior patterns. The SOC team triaged the alerts, validated indicators using threat intelligence, and initiated containment by isolating affected systems and blocking malicious IPs. SOAR automation significantly reduced response time by executing predefined playbooks for containment and notification. Adversary emulation techniques were used to validate detection mechanisms and identify gaps. Post-incident analysis revealed that weak configurations and delayed detection contributed to the attack's success. Root Cause Analysis identified a lack of monitoring and insufficient rule tuning as primary issues. Metrics analysis showed an MTTD of 2 hours and a MTTR of 4 hours, indicating moderate efficiency but room for improvement. Recommendations include enhancing detection rules, implementing continuous monitoring, and increasing user awareness to prevent phishing attacks. This project demonstrates a complete SOC workflow, including detection, response, automation, and continuous improvement, aligning with industry best practices.

Complete attack simulation using Metasploit, detection via Wazuh, response via CrowdSec, automation using SOAR, RCA, and reporting.

Timestamp	Source IP	Alert	MITRE
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2025-08-18	192.168.1.102	Samba exploit	T1210

Conclusion

This project demonstrates practical SOC expertise including threat hunting, automation, incident response, and reporting. It reflects real-world readiness for cybersecurity roles.